

[illegible]

domain. FFT is an algorithm to compute DFT in a fast way. It is generally performed using decimation-in-time (DIT) approach. Here we give a brief introduction to DIT approach and implementation of the same in C++.

flow graph in Fig. 3 shows a useful representation of a system of equations.

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

$N=2^m$
where 'm' is a positive integer.

Here we have taken $m=3$, so $N=8$. The signal

Software program

The software, written in C++, is compiled using Turbo C++ Version 3.0. It was tested on Windows XP SP3 machine. Run the FFT.exe file and enter each signal element of an array followed by pressing Return/Enter key. Up to eight input elements can be entered.

Fig. 4 shows the screenshot of program output. Here the coefficient of input 'x' is (1, 3, 5, 7, 9, 11, 13, 15). The output is given in (A, B) format, which indicates 'A' is real and 'B' imaginary component of the complex number. In order to verify the correctness of our code, we calculated FFT of the same sequence (1, 3, 5, 7, 9, 11, 13, 15) in MATLAB as shown in Fig. 5.

It can be seen that MATLAB outputs the same FFT coefficients as our C++ code, which proves the correctness of our code.

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```
#include<iostream.h>
#include<conio.h>
#include<math.h>
#include<dosh.h>
#include<complex.h>
#define pi 3.141592654

void main()
{ clrscr();
int i,x,y,j,k,l;
float o,r,t;
complex p,q,r,s,w,m,n,alpha,beta,gamma[10],g
gamma2[10];
complex v[i][j],b[i][o],[a][l],[c][t],[d][l],[gamma]
[O],[w][t],[z][t];
cout<<"enter the elements of the array "<<endl;
for(i=0;i<j;i++)
cin>>a[i];
cout<<endl<<endl<<endl;
cout<<"Enter b["<<t"<<endl<<endl;
for(i=0;i<l+i++)
{ cout<<a[i]<<endl;
for(i=0;i<3;i++)
b[i]=a[i-j];
}
for(i=4,j=3;i<7,j+=0,i=i+1,j=j-1)
b[i]=a[i-j];
for(i=0;i<1+j++)
```